

Enhanced solubility and stability of beta-carotene in choline chloride-based natural deep eutectic solvents: insights from molecular dynamics simulations and physicochemical characterization

Yaremis Meriño-Cabrera^{a,b,*}, Ivana Arrieta-Barba^{a,b}, Luisa Florez-Vargas^b,
Tiana Serna Fadul^b, Jorge Figueroa-Florez^c, José Severiche-Castro^b, Jairo Salcedo Mendoza^b

^a Programa Orquideas Minciencias 2024: Mujeres en la Ciencia, Entidad ejecutora: Universidad de Sucre, Sincelejo, Colombia

^b Programa de Ingeniería Agroindustrial, Grupo de investigación en procesos agroindustriales y desarrollo sostenible (PADES), Universidad de Sucre, Sincelejo, Colombia

^c Programa de Ingeniería Agroindustrial, Facultad de Ingeniería y Administración, Universidad Nacional de Colombia, Palmira, Valle del Cauca, Colombia

ARTICLE INFO

Keywords:

Carotenoids
Computational modeling
Solvation
Green solvents

ABSTRACT

β -Carotene is a lipophilic carotenoid with well-established nutritional and pharmacological benefits. However, its industrial use is limited by poor water solubility and high oxidative instability. Natural deep eutectic solvents (NADES), especially those based on choline chloride, have emerged as promising green alternatives for bioactive extraction. Nevertheless, their capacity to solubilize and stabilize highly hydrophobic molecules like β -carotene remains insufficiently explored. This study integrates atomistic molecular dynamics (MD) simulations with experimental physicochemical analyses to evaluate the solubility, dispersion, and stability of β -carotene in choline chloride-based NADES. Systems containing β -carotene in methanol–water, dichloromethane (DCM), and two NADES—choline chloride–malic acid (ChoCl–Mal) and choline chloride–oxalic acid (ChoCl–Oxa)—were examined using MD simulations by analyzing solvent-accessible surface area (SASA), radius of gyration (Rg), mean squared displacement (MSD), and radial distribution functions (RDF). Complementary experiments—including FTIR and UV–Vis spectroscopy, rheology, and zeta potential measurements—validated computational predictions and assessed solute–solvent interactions. The ChoCl–Mal system demonstrated the most favorable overall performance: it enhanced surface exposure, maintained stable dispersion, exhibited Newtonian flow, and provided moderate colloidal stability. Spectroscopic data confirmed chromophore preservation and specific molecular interactions, while rheological and electrokinetic analyses supported improved physicochemical protection. Although DCM showed superior solubilization, its low colloidal stability limits long-term applicability. These findings identify ChoCl–Mal as a technically robust and environmentally sustainable solvent for β -carotene extraction and formulation, offering a viable alternative to conventional solvents in food and pharmaceutical contexts.

1. Introduction

The growing demand for natural bioactive compounds in the food and pharmaceutical industries has intensified the search for effective and sustainable extraction methods. Carotenoids, such as β -carotene, are natural pigments with significant health benefits due to their antioxidant properties and provitamin A activity [1]. However, their low solubility and susceptibility to oxidative degradation limit their practical application in food and pharmaceutical products [2]. Conventional organic solvents have been widely used for β -carotene extraction, but

they pose toxicity concerns and environmental challenges, prompting the search for green alternatives [3].

The efficient extraction of natural products from biomass using environmentally friendly solvents is a crucial area of interest in the food, pharmaceutical, and cosmetic industries [4]. In this context, natural deep eutectic solvents (NADESs) have emerged as promising alternatives due to their non-toxic nature, tunable physicochemical properties, and high extraction efficiency for bioactive compounds [5].

NADESs belong to a new generation of green solvents, sharing certain characteristics with ionic liquids (ILs). However, ILs have

* Corresponding author at: Universidad de Sucre, Sincelejo, Colombia.

E-mail address: yaremis.cabrera@ufv.br (Y. Meriño-Cabrera).

<https://doi.org/10.1016/j.molliq.2025.128251>

Received 22 April 2025; Received in revised form 29 July 2025; Accepted 30 July 2025

Available online 5 August 2025

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notable drawbacks, including poor biodegradability, potential toxicity, and environmental hazards that may affect ecosystems and living organisms [6,7]. Additionally, the high production costs of ILs limit their economic feasibility for large-scale application [8]. In contrast, NADESs offer multiple advantages, making them particularly suitable for extraction processes: they can be prepared using simple, cost-effective methods [9], exhibit high solubility for a wide range of compounds [10], and are composed of biodegradable components, aligning with the principles of sustainable development [5,11].

Structurally, NADESs are formed by mixing a hydrogen bond acceptor (HBA) with a hydrogen bond donor (HBD) in specific molar ratios, resulting in a stable eutectic mixture with unique solubilization properties [3,12]. A common HBA in NADES formulations is choline chloride (ChCl), while HBDs include organic acids, sugars, and polyols, which modulate solvent polarity and viscosity [13]. These interactions significantly influence the solubility of various bioactive compounds. For instance, the solubility of β -carotene in NADES is driven by hydrogen bonding, van der Waals interactions, and π - π stacking with solvent molecules [14].

While NADESs have demonstrated high potential for the extraction of hydrophobic bioactives, the molecular-level mechanisms that govern their solubility and stabilization remain insufficiently understood. In particular, the efficiency of β -carotene extraction from plant matrices is closely linked to how well the compound is solvated and stabilized within the solvent environment [15]. However, experimental approaches alone often fall short in capturing the dynamic and structural complexities of solute-solvent interactions. In this context, molecular dynamics (MD) simulations offer a powerful computational framework to explore these interactions at the atomic scale, enabling detailed assessments of solvation behavior, conformational stability, and specific binding events that drive β -carotene dispersion and stabilization in NADES [16]. To validate these computational predictions and capture macroscopic behaviors such as colloidal stability and vibrational signatures, experimental techniques such as FTIR, UV-Vis spectroscopy, rheological profiling, and zeta potential analysis are essential complementary tools.

NADESs composed of choline chloride and various hydrogen bond donors have demonstrated potential in extracting hydrophilic and moderately hydrophobic compounds [17]. However, their suitability for β -carotene remains largely unexplored. Although MD simulations provide valuable atomistic insights into solvation and stabilization phenomena, they are most powerful when combined with experimental data that confirm macroscopic behaviors. In this study, we complement simulation results with detailed experimental characterization to correlate molecular interactions with measurable physicochemical properties. The application of MD simulations in this context helps guide the optimization of extraction conditions and enhances the efficiency of β -carotene recovery [18].

Although previous works have explored NADES formulations for hydrophilic compounds, their effectiveness in stabilizing lipophilic carotenoids such as β -carotene has not been comprehensively evaluated through atomistic simulations combined with experimental validation. In recent years, various theoretical approaches have been developed to model solubility and molecular interactions in deep eutectic solvents. Among these, the conductor-like screening model for real solvents (COSMO-RS) has been widely applied to predict the solvation behavior and thermodynamic properties of bioactive compounds in NADES, including poorly water-soluble molecules such as pesticides [19], flavonoids like rutin [20], and diverse pharmaceutical agents such as ibuprofen, ketoprofen, and sulfonamides [21,22]. These approaches offer valuable insights into thermodynamic properties and solvent screening but are limited in capturing time-dependent dynamics and conformational flexibility.

More recently, the integration of molecular modeling with machine learning techniques has gained momentum as a powerful tool to predict solubility, optimize solvent design, and accelerate green chemistry

applications. In this broader context, molecular dynamics (MD) simulations provide atomistic resolution and temporal evolution of solute-solvent systems, complementing thermodynamic models and data-driven methods. Thus, the present work contributes to this evolving field by combining MD simulations with experimental validation to study β -carotene solubilization in NADES, aiming to bridge molecular-level interactions with macroscopic solvent behavior.

Therefore, the objective of this study is to comprehensively evaluate the solubility, structural stability, and molecular interactions of β -carotene in choline chloride-based NADES by combining atomistic molecular dynamics simulations with experimental physicochemical characterization (FTIR, UV-Vis spectroscopy, rheological analysis, and zeta potential). This integrative approach seeks to identify the most effective eutectic solvent system for β -carotene recovery, elucidate the mechanisms underlying solute stabilization, and support their practical implementation in advanced extraction systems, colloidal formulations, and delivery platforms for lipophilic bioactives. Ultimately, the findings aim to provide a rational basis for the design of green, high-performance solvents applicable to carotenoid extraction in food and pharmaceutical applications.

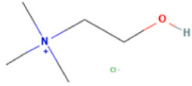
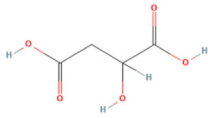
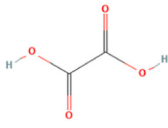
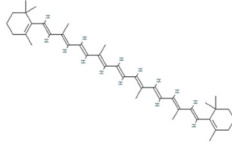
2. Materials and methods

2.1. Molecular structure preparation

The initial molecular structures of the bioactive compound β -carotene and the solvents used in this study (systems S3 to S5, Table 1) were generated using Avogadro software [23], a widely adopted tool for molecular modeling and visualization. A preliminary conformational analysis of β -carotene was conducted to identify its lowest-energy structure. Multiple conformers were generated by rotating key single bonds, followed by pre-optimization using the semiempirical PM6 method [24] and subsequent refinement via density functional theory (DFT). The lowest-energy conformer was selected for further calculations, ensuring that the geometry employed reflects a stable and representative configuration under standard conditions.

Table 1 summarizes the chemical structures of the components used in the formulation of the solvents, including choline chloride, malic acid, oxalic acid, water, dichloromethane, and methanol. Solvent S1 consists of pure water, S2 is a binary water-methanol mixture, S3 is pure

Table 1
Structure and chemical formula of the eutectic solvent molecules and bioactive compound studied.

Solvents and chemical formula	Chemical structure
Choline chloride $C_5H_{14}ClNO$	
Malic acid $C_4H_6O_5$	
Oxalic acid $C_2H_2O_4$	
β -carotene $C_{40}H_{56}$	

dichloromethane, S4 comprises choline chloride, malic acid, and water, and S5 contains choline chloride, oxalic acid, and water.

2.1.1. Geometric optimization and parameterization

To ensure the structural and energetic accuracy of the studied molecules, a geometry optimization was carried out using density functional theory (DFT). The B3LYP functional [24] was employed in combination with the 6-311G(d,p) basis set [25], a level of theory widely validated for the accurate description of organic systems. Calculations were performed using the Gaussian 09 software package. To confirm that the optimized structures correspond to true minima on the potential energy surface, vibrational IR spectra were computed, confirming the absence of imaginary frequencies for both β -carotene and the solvents considered (Supplementary Tables 1–4).

The molecular topologies of the eutectic solvents and β -carotene were generated using the CHARMM-GUI platform [26], a specialized tool for preparing systems for molecular dynamics (MD) simulations. Molecular parameters were derived from the CHARMM36 force field [27], enabling an accurate representation of the interatomic interactions within the simulated system.

2.2. Molecular dynamics simulations

Molecular dynamics (MD) simulations were performed to evaluate the interaction between the bioactive compound β -carotene and various solvent systems. Cubic simulation boxes measuring $6 \times 6 \times 6 \text{ nm}^3$ were constructed, and each system was solvated using the TIP3P water model. Four systems were analyzed in total: S1 consisted solely of water; S2 contained a mixture of water and methanol; S3 consisted solely dichloromethane, S4 was a natural deep eutectic solvent (NADES) composed of choline chloride as a hydrogen bond acceptor (HBA), malic acid as a hydrogen bond donor (HBD), and water; and S5 was another NADES formed by choline chloride (HBA), oxalic acid (HBD), and water.

The NADES systems (S4 and S5) were prepared using a 1:1 M ratio between the HBA and HBD components. Additionally, each NADES system included 15 β -carotene molecules. For S4, the system contained 350 molecules of choline chloride, 340 molecules of malic acid, and 1600 water molecules. In S5, 330 molecules of choline chloride, 510 molecules of oxalic acid, and 1400 water molecules were used.

Initial configurations for each system were generated using the PACKMOL software, which allows for the random distribution of molecules within the simulation box while avoiding overlaps. Energy minimization was then carried out using the Steepest Descent algorithm, with a convergence threshold of 1000 kJ/mol-nm. Subsequently, a molecular dynamics simulation was run for 500,000 steps with a time step of 0.001 nm.

The molecular dynamics simulations were performed using the GROMACS software suite, following a three-phase protocol: energy minimization, equilibration, and production runs, with respective durations of 45 ns, 60 ns, and 100 ns. During these stages, the system temperature was maintained using the V-rescale thermostat, and the LINCS algorithm was applied to constrain bonds involving hydrogen atoms, ensuring a constant temperature of 310.15 K.

Finally, the resulting trajectories were analyzed using tools from the GROMACS package and visualized with Visual Molecular Dynamics (VMD) software.

2.3. Visual analysis of solubility and molecular interactions

Representative visualizations of the simulation trajectories were generated by capturing snapshots at key moments of the process (initial and final). To study the organization of the solvent around β -carotene, electron density maps were constructed using the VolMap module in VMD. These qualitative analyses enabled the examination of the uniformity of solvent coverage, the presence of molecular aggregation, and the formation of structured microenvironments or solvent-depleted

regions.

To characterize specific interactions—such as hydrogen bonds, hydrophobic stacking, and functional associations—a proximity analysis was performed, considering interactions within 3.5 Å between β -carotene and relevant solvent functional groups. The identification of stable interaction geometries was carried out through a combined manual and automated inspection of the molecular trajectories. These interactions were further analyzed using Discovery Studio software, allowing for comparisons across the different simulated systems.

2.4. Structural and dynamic analysis

The structural and dynamic analysis of β -carotene was conducted using the analysis modules of VMD (version 1.9.4), complemented with custom Python scripts. Several key structural parameters were evaluated, including:

- Solvent-Accessible Surface Area (SASA): Calculated using the Lee & Richards algorithm implemented in VMD and averaged over the final 100 ns of simulation for each system, with the aim of quantifying the surface exposure of β -carotene to the surrounding environment.
- Radius of Gyration (Rg): Used as a measure of the structural compactness of the molecule.
- Mean Squared Displacement (MSD): Employed to estimate the mobility of β -carotene over time, facilitating comparison of its diffusive dynamics across different solvents.
- Radial Distribution Functions (RDF): Calculated between selected atoms of β -carotene and solvent components to characterize the local organization of the molecular environment and identify specific interaction patterns.

2.5. Functional and physicochemical insights into β -carotene solubilization in NADES

2.5.1. FTIR spectroscopy

Fourier-transform infrared (FTIR) spectra of β -carotene in the four (S2–S5) solvent systems were recorded using an FTIR spectrometer equipped with an attenuated total reflectance (ATR) accessory, following the protocol described by [28] with minor modifications. Spectra were acquired over the range of 4000–650 cm^{-1} with a resolution of 2 cm^{-1} , averaging 64 scans per sample. Approximately 2 mg of β -carotene were placed directly onto the surface of the ATR crystal for spectral acquisition. All spectra were baseline-corrected and normalized to enable comparison of vibrational modes. Characteristic absorption bands associated with β -carotene and potential solvent interactions were identified and annotated.

2.5.2. UV-vis absorption

UV-Vis spectra were acquired using a dual-beam spectrophotometer (Perkin-Elmer, Lambda 365, USA) over the wavelength range of 300–600 nm, with a scan rate of 600 nm/min and a data interval of 1.0 nm. β -Carotene was dissolved in the four solvent systems at equivalent concentrations ($\sim 10 \mu\text{g/mL}$). Quartz cuvettes with a 1.0 cm path length were employed, and solvent blanks were automatically subtracted. The resulting absorbance spectra were analyzed to evaluate the solubility, dispersion quality, and spectral integrity of the β -carotene chromophore, with particular attention to the vibronic fine structure in the 450–500 nm region.

2.5.3. Rheological behavior

The rheological behavior of β -carotene dispersions was evaluated through dynamic rotational tests using a modular compact rheometer (MCR302, Anton Paar, Austria) equipped with a Peltier system and a thermostatic bath for precise temperature control. Flow curves were obtained using a concentric cylinder geometry over a shear rate range of 0.1 to 100 s^{-1} at a constant temperature of 25 °C, following the protocol

described by [28] with minor modifications. Shear rate sweeps were conducted in both ascending and descending modes to identify and minimize potential thixotropic effects. Rheological parameters were determined by fitting the flow data to the Newtonian, Power Law, and Herschel–Bulkley models, selecting the best-fitting model based on the coefficient of determination (R^2), root mean square error (RMSE), and the associated p -value.

2.5.4. Zeta potential analysis

The zeta potential (ζ) was determined by dynamic light scattering (DLS) using a Zetasizer instrument (model ZS90, Malvern Instruments, UK) equipped with a high-concentration cell and electrophoretic mobility capabilities, following the methodology described by [29]. β -Carotene dispersions were diluted at a 1:10 ratio with their respective solvents to minimize multiple scattering effects. A refractive index of 1.56 was assumed for β -carotene in dispersion for the zeta potential calculations. Measurements were conducted in triplicate across independent runs at a controlled temperature of 25 °C, following an equilibration time of 120 s. The resulting data were used to assess the colloidal stability of the dispersions based on the electrostatic repulsion between particles. Additionally, digital images of each sample were captured before and after carotenoid incorporation to visually document the dispersion state.

3. Result and discussion

In order to evaluate the influence of the solvent environment on the interactions of beta-carotene, electrostatic, van der Waals, and total energies were calculated for the four systems studied, including pure water, a methanol–water mixture, and two deep eutectic solvents (DES) based on choline chloride combined with organic acids (malic acid or oxalic acid) in aqueous solution. The results obtained are summarized in Table 2.

The interaction energies of β -carotene vary significantly across the different solvent systems analyzed (Table 2). The least stabilizing environment was observed in the pure water system (S1), which exhibited the highest total interaction energy (−6106.29 kcal/mol). In contrast, the methanol–water mixture (S2) showed a substantial decrease in total energy (−110,685.00 kcal/mol), indicating improved solute–solvent interactions likely driven by enhanced solvation capacity due to the presence of methanol.

In terms of interaction energies, the dichloromethane (DCM) system (S3) exhibited a distinct profile compared to the other solvent systems. DCM showed a relatively weak electrostatic interaction with β -carotene (−378.53 kcal/mol), which aligns with its low polarity and limited ability to engage in directional electrostatic interactions. In contrast, the van der Waals contribution was considerably stronger (−3773.01 kcal/mol), suggesting that dispersion forces dominate the stabilization mechanism in this system. The total interaction energy (−30,913.70

kcal/mol) was notably more negative than that of water but still significantly less stabilizing than the methanol–water mixture and the NADES systems, highlighting its intermediate performance as a solvation environment for β -carotene.

The most favorable total interaction energies were observed in the deep eutectic solvent (DES) systems. System S4 (choline chloride + malic acid + water) exhibited the lowest total energy (−405,139.00 kcal/mol), followed by S5 (choline chloride + oxalic acid + water) with −259,485.00 kcal/mol. These results confirm the superior stabilizing effect of NADES, driven primarily by strong electrostatic interactions and moderate van der Waals contributions.

Regarding the electrostatic component, the most negative values were found in S5 (−228,842.00 kcal/mol), followed by S4 (−166,066.00 kcal/mol), S2 (−112,193.00 kcal/mol), and S1 (−10,063.30 kcal/mol) (Table 2). As expected, DCM (S3) presented the weakest electrostatic interaction (−378.53 kcal/mol). For van der Waals interactions, values ranged from −13,467.20 kcal/mol in S2 to 3452.75 kcal/mol in S5. The DCM system again showed a relevant van der Waals contribution (−3773.01 kcal/mol) (Table 2), although not as high as in the methanol–water mixture.

3.1. Visual analysis of solubility and molecular interactions

A qualitative visual analysis of β -carotene distribution in the different solvent media was performed using representative snapshots extracted from the molecular dynamics simulations. The figures are organized into four blocks (Figs. 1–3), corresponding to the systems under study (S2–S5), along with a comparative set of molecular interactions (Fig. 4).

In System S2, consisting of a water–methanol mixture, the initial configuration (Fig. 1A) shows β -carotene spatially separated from the rest of the system. In the complete system visualization (Fig. 1B), initial aggregation of the solvent around the solute is observed, although no homogeneous coverage is evident. After 100 ns of simulation, β -carotene remains in a partially clustered configuration (Fig. 1C), while the solvent isosurfaces (Fig. 1D) reveal a non-uniform distribution in its immediate environment. These results suggest incomplete dissolution of β -carotene, with the formation of localized microenvironments and structural voids within the solvent mixture.

In System S3, an initially dispersed arrangement of β -carotene is observed within the simulated volume (Fig. 2A), suggesting a lower tendency toward aggregation from the beginning of the simulation. Throughout the process, the solute maintains an open and distributed configuration, as evidenced by the molecular representations (Fig. 2B and C), where multiple β -carotene molecules occupy distinct regions of the system. The visualization of the solvent electron density (Fig. 2D) reveals continuous and well-distributed coverage around the solute, indicating more effective interactions between β -carotene and the NADES components. This arrangement suggests a greater capacity of System S3 to stabilize and solubilize β -carotene compared to the previously evaluated systems.

In System S4, a more cohesive configuration of β -carotene is observed from the beginning of the simulation (Fig. 3A and B) compared to that recorded in System S3 (Fig. 2). The molecular visualizations reveal a more densely organized environment around the solute, with a compact distribution of the system's molecules. The electron density visualization (Fig. 3D) confirms this trend, showing a well-defined packing around β -carotene. This organization reflects a more localized and efficient solvation. The orientation of β -carotene suggests a structured accommodation within the microenvironment generated by the NADES, indicating greater compatibility and affinity between the solute and the solvent medium.

The key molecular interactions between β -carotene and the components of the solvent media in the four systems analyzed reveal distinct solvation behaviors. In the methanol–water mixture, β -carotene forms a few weak interactions with methanol molecules (Fig. 4A), but these do

Table 2

Interaction energies of beta-carotene in different solvent environments. Electrostatic, van der Waals, and total energy (kcal/mol) are reported for each system. The solvents include pure water, a methanol–water mixture, dichloromethane and deep eutectic solvents (DES) based on choline chloride combined with either malic acid or oxalic acid, all in aqueous solution.

System	Solvent	Electrostatic Energy	Van der Waals Energy	Total Energy
S1	Water	−10,063.3	1070.28	−6106.29
S2	Methanol + Water	−112,193	−13,467.2	−110,685
S3	Dichloromethane	−378.53	−3773.01	−30,913.7
S4	Choline Chloride + Malic Acid + Water	−166,066	664.121	−405,139
S5	Choline Chloride + Oxalic Acid + Water	−228,842	3452.75	−259,485

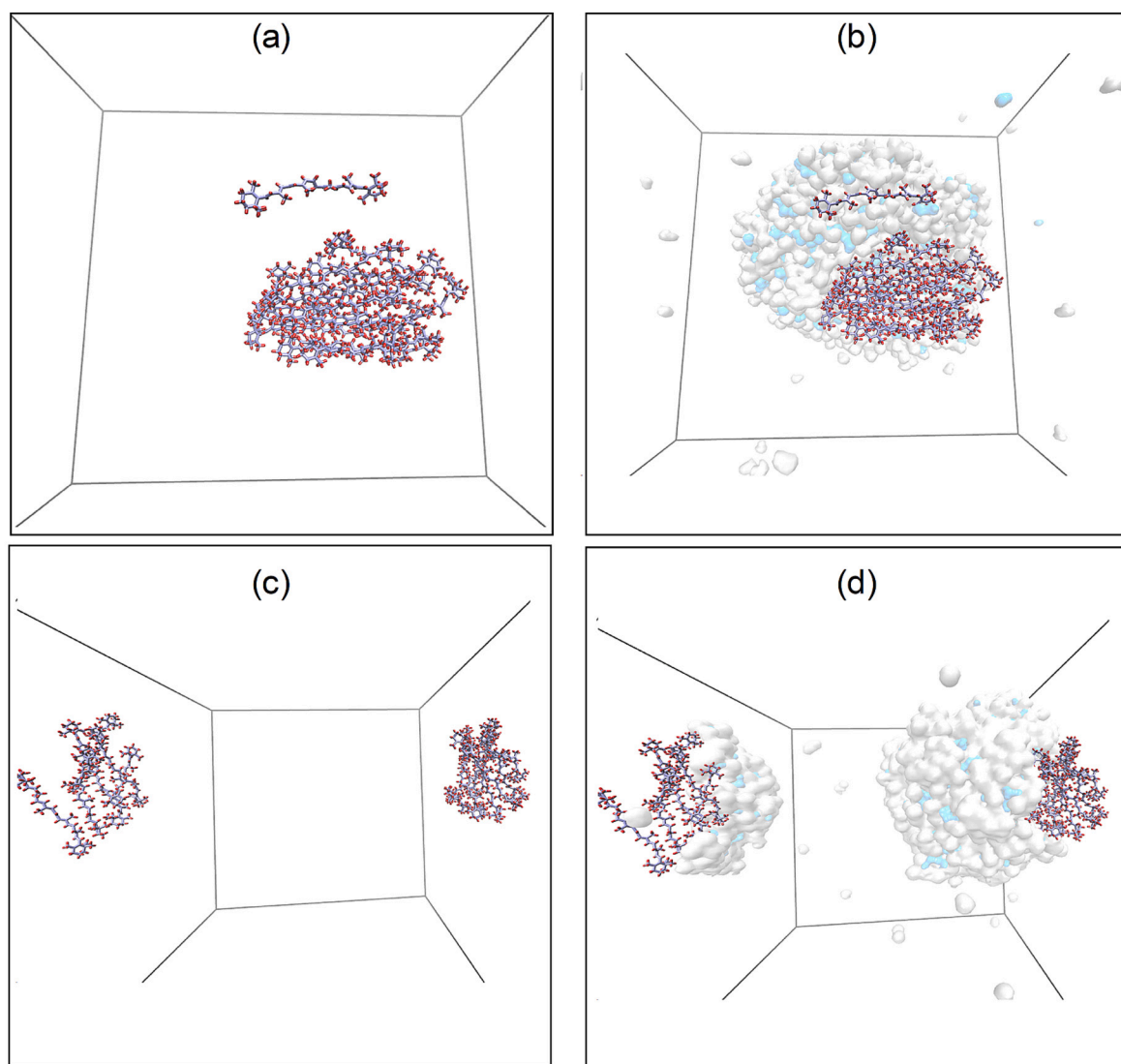


Fig. 1. Visual diagram of beta-carotene dissolution in System 2 (S2): a mixture of water (H_2O) and methanol (CH_3OH). Panels (A) and (B) correspond to the initial state (0 ns), while (C) and (D) show the system after 100 ns of simulation. In (A) and (C), only the beta-carotene molecules are displayed, whereas in (B) and (D), the full system is shown, including the solvent (water and methanol), with isosurfaces highlighting the interaction between the solute and the surrounding medium.

not result in a stable or structured solvation environment. In the dichloromethane system (Fig. 4B), β -carotene engages primarily in alkyl-alkyl interactions with surrounding DCM molecules. These nonpolar van der Waals contacts are dispersed along the polyene chain of the solute, reflecting a lack of directional or specific hydrogen bonding. As a result, the solvation environment appears less ordered and more dynamic, consistent with the low polarity and moderate stabilizing capacity observed in the energy analysis.

In contrast, in the choline chloride-malic acid NADES, β -carotene establishes multiple interactions with malic acid molecules (Fig. 4C), including hydrogen bonding with functional groups of the solvent, suggesting a stronger affinity of the solute for this eutectic medium. Similarly, in the system containing choline chloride and oxalic acid, hydrogen bonds are also observed between β -carotene and oxalic acid (Fig. 4D), with a more compact and organized geometry. This indicates enhanced structural stability and more localized interaction within the microenvironment generated by this NADES formulation.

3.2. Structural and dynamic analysis

The analysis of radial distribution functions (RDF) enabled the

quantification of the structural organization of the solvation environment around β -carotene across the different simulated systems. In the aqueous system (S1), a sharp and pronounced peak is observed in the $g(r)$ curve at approximately 0.85 nm, with a maximum value close to 55 (Fig. 5). This indicates a strong accumulation of water molecules around β -carotene at short range. The high local density suggests that β -carotene induces significant reorganization of the water network, generating a highly structured solvation shell—likely driven by the hydrophobic nature of the molecule, which promotes tightly packed solvent arrangement.

In the methanol-water mixture (S2), the RDF profiles reveal a more diffuse interaction pattern. The RDF between β -carotene and water displays a broader, less intense peak, with a $g(r)$ maximum below 8, while the interaction with methanol is even weaker (Fig. 5). These results indicate a less structured solvent organization around β -carotene, consistent with the formation of partially solvated aggregates (Fig. 1).

The dichloromethane system (S3) shows an intermediate behavior between structured and disordered solvation. A moderate peak is observed at around 0.85 nm, with a $g(r)$ maximum of approximately 22, indicating a localized but weaker accumulation of DCM molecules near β -carotene compared to water. The RDF profile suggests that DCM forms

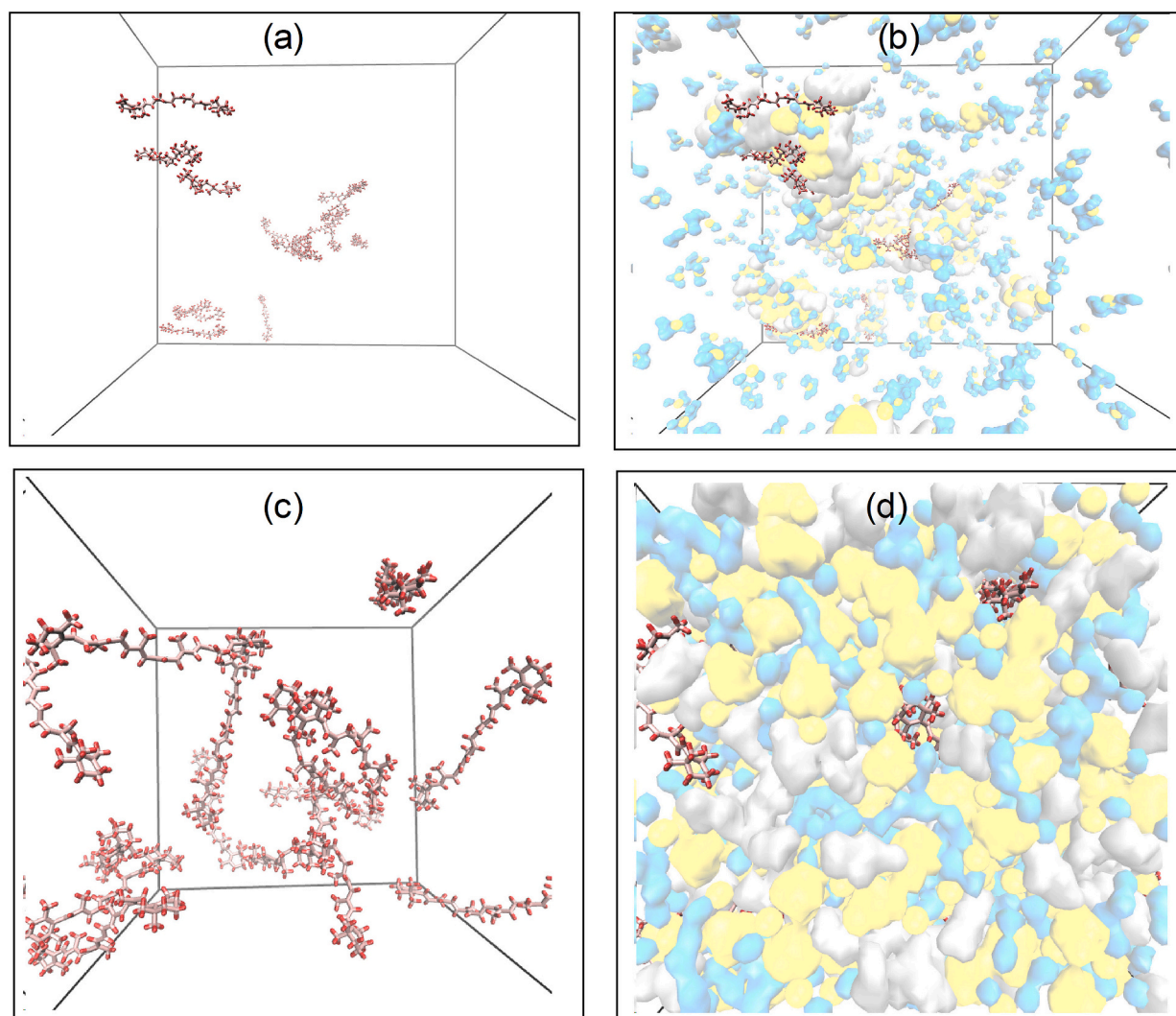


Fig. 2. Visual diagram of beta-carotene dissolution in System 3 (S4): a mixture of choline chloride ($C_5H_{14}ClNO$), malic acid ($C_4H_6O_5$), and water (H_2O). Panels (A) and (B) correspond to the initial state (0 ns), while (C) and (D) show the system at 100 ns. In (A) and (C), only the beta-carotene molecules are displayed, whereas in (B) and (D), the full system is shown, including the solvent (choline chloride, malic acid, and water), with isosurfaces highlighting the interaction between the solute and the surrounding medium.

a relatively compact but less organized solvation layer, dominated by nonpolar van der Waals interactions. This is consistent with the nature of DCM as a nonpolar, aprotic solvent that interacts with β -carotene mainly through dispersion forces without inducing significant restructuring of the surrounding medium.

In systems S4 (ChoCl–Mal) and S5 (ChoCl–Oxa) (Fig. 5), the RDF values remain relatively low and close to unity (1), indicating an almost homogeneous distribution of solvent components around the β -carotene molecule. The absence of pronounced peaks suggests a lack of directed interactions or strong local structuring near the solute.

In S4, the RDF profiles between β -carotene and all three components (water, malic acid, and choline chloride) remain flat, which is indicative of effective solute dispersion within the DES medium. A similar behavior is observed in S5, although slight variations are noted in the $g(r)$ curves—particularly in the interaction with oxalic acid (Bet–Oxa)—which may be attributed to its smaller molecular size and higher propensity to form hydrogen bonds.

To assess the structural conformation and surface accessibility of β -carotene in different solvent environments, the solvent-accessible surface area (SASA) and radius of gyration (R_g) were computed over 100 ns of molecular dynamics simulations (Fig. 6). The temporal evolution of the SASA reveals pronounced differences among the systems. In

System S1, the SASA remains close to 0 nm² throughout the entire simulation, indicating complete insolubility and pronounced hydrophobic stacking. This behavior likely reflects molecular aggregation or structural collapse, severely limiting the exposure of β -carotene to the solvent environment (Fig. 6).

In System S2, the SASA remains constant at approximately 60 nm², suggesting a slight improvement in surface exposure compared to the purely aqueous system. However, this level of accessibility remains low, which is consistent with the low mobility observed in the MSD analysis and the potential formation of partially exposed aggregates (Fig. 6).

In System S3 (DCM), a sharp increase in SASA is observed during the first 20 ns, stabilizing around 80–90 nm². This moderate surface accessibility, higher than that of water-based systems but lower than in NADES, reflects the ability of DCM to solvate β -carotene through non-polar dispersion interactions. Although DCM lacks hydrogen bonding capacity, its compatibility with the hydrophobic polyene chain of β -carotene facilitates a more extended and solvated structure than polar solvents like water or methanol.

In contrast, the NADES-based systems exhibit distinct behavior. In S4 (ChoCl–Mal), SASA remains high, with average values around 140–150 nm², while in S5 (ChoCl–Oxa), it is slightly lower, ranging between 120 and 130 nm². This increase in surface accessibility in DES media

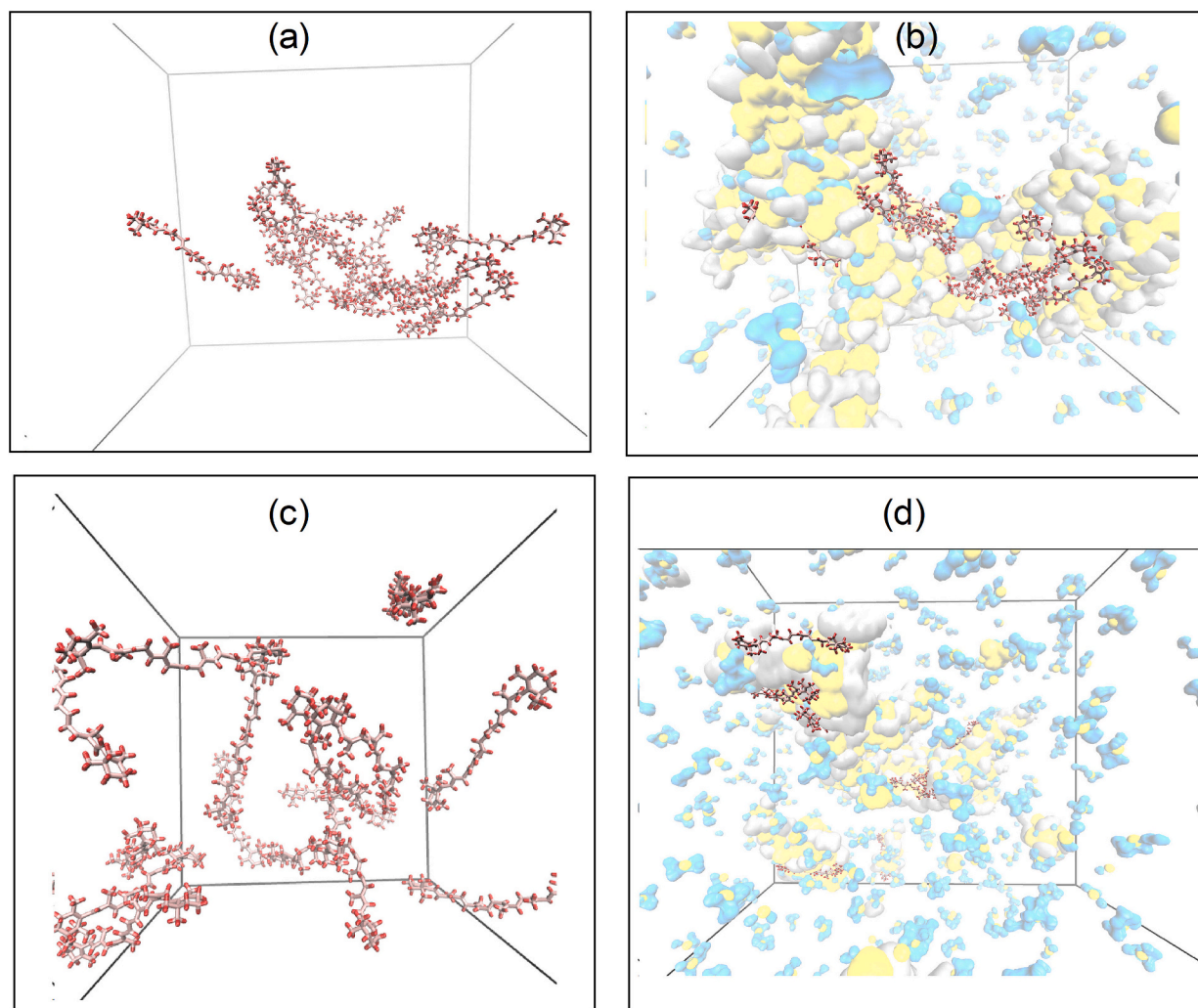


Fig. 3. Visual diagram of beta-carotene dissolution in System 4 (S5): a mixture of choline chloride ($C_5H_{14}ClNO$), oxalic acid ($C_2H_2O_4$), and water (H_2O). Panels (A) and (B) correspond to the initial state (0 ns), while (C) and (D) show the system at 100 ns. In (A) and (C), only the beta-carotene molecules are displayed, whereas in (B) and (D), the full system is shown, including the solvent (choline chloride, oxalic acid, and water), with isosurfaces highlighting the interaction between the solute and the surrounding medium.

suggests that β -carotene is more exposed to the surrounding environment, possibly as a result of more efficient solvation and reduced aggregation, favored by specific interactions with the components of the NADES.

On the other hand, the analysis of the radius of gyration (R_g) provided complementary information on the structural compactness of β -carotene. In System S1, R_g values remained consistently low (~ 0.9 nm), confirming a highly collapsed or compact structure with minimal exposure to the surrounding medium (Fig. 6). In contrast, System S2 exhibited large fluctuations in R_g , with peaks reaching up to 8–9 nm, indicating high structural instability and possible transitions between collapsed and partially extended states. This is consistent with an environment that fails to effectively stabilize the conformation of β -carotene (Fig. 6).

In System S3 (DCM), R_g increases continuously throughout the simulation, reaching values above 15 nm. This progressive expansion suggests a highly extended and flexible β -carotene conformation in DCM, driven by the absence of polar interactions and the prevalence of weak dispersion forces.

In System S4 (ChoCl-Mal), the R_g stabilized around 3–4 nm, whereas in S5 (ChoCl-Oxa) the average value was lower, approximately 2 nm. These differences reflect a more extended β -carotene structure in ChoCl-Mal compared to ChoCl-Oxa, which correlates with the higher SASA

observed in System S4. These observations suggest that the environment provided by ChoCl-Mal enables more effective solvation, resulting in a more open molecular structure (Fig. 6).

The analysis of the mean squared displacement (MSD) of β -carotene in the different solvent environments is presented in Fig. 7. The results allow for a comparison of the molecule's relative mobility as a function of the physicochemical characteristics of each medium. In System S1, high mobility is observed, with a progressive increase in MSD reaching values above 1400 nm^2 at 100 ns. This behavior indicates significant diffusion of β -carotene in the aqueous medium, presumably facilitated by the low viscosity of the solvent and the limited specific interactions between the solute and water molecules (Fig. 7).

In contrast, in System S2, the MSD remains nearly constant throughout the simulation, with values close to 10 nm^2 , indicating extremely restricted mobility of β -carotene in this mixture. This result suggests the formation of stable aggregates or strong interactions between β -carotene and the solvation environment that hinder its diffusion (Fig. 7).

In System S3 (DCM), β -carotene exhibits the highest MSD values among all the systems evaluated, exceeding 3600 nm^2 at the end of the 100 ns simulation. This pronounced mobility is consistent with the low polarity and low viscosity of DCM, which promotes weak, non-directional dispersion interactions and enables the free diffusion of the

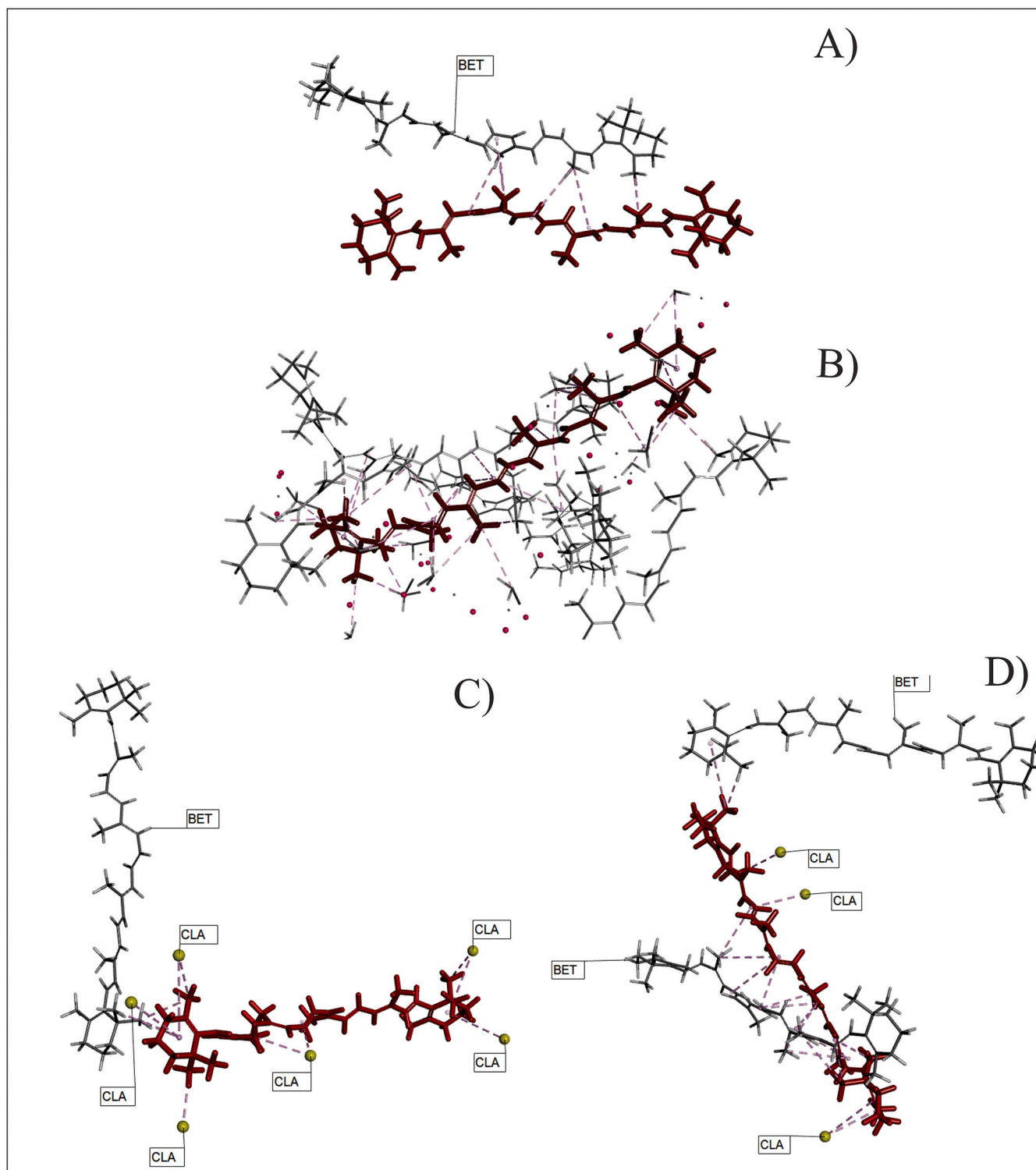


Fig. 4. Comparison of interactions between beta-carotene and solvents in different systems. (A) S2: mixture of water (H_2O) and methanol (CH_3OH), (B) S3: dichloromethane (DCM , CH_2Cl_2), (C) S4: mixture of choline chloride ($\text{C}_5\text{H}_{14}\text{ClNO}$), malic acid ($\text{C}_4\text{H}_6\text{O}_5$), and water (H_2O), (D) S5: mixture of choline chloride ($\text{C}_5\text{H}_{14}\text{ClNO}$), oxalic acid ($\text{C}_2\text{H}_2\text{O}_4$), and water (H_2O). The structures show possible interactions between beta-carotene (dark red sticks) and solvent components, highlighting the influence of the chemical environment on its spatial organization. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

hydrophobic β -carotene molecule. The absence of hydrogen bonding or strong dipolar interactions allows the solute to explore a wide conformational and translational space, supporting a highly dynamic environment.

For the NADES systems, S4 and S5 show significantly lower MSD

values than the pure aqueous system, confirming restricted mobility of β -carotene in these more viscous media (Fig. 7). However, a clear difference is evident between the two DES: System S5 reaches MSD values of approximately 1.2 nm^2 , while S4 remains below 0.6 nm^2 over the same time interval (Fig. 7). This suggests that the choline chloride/

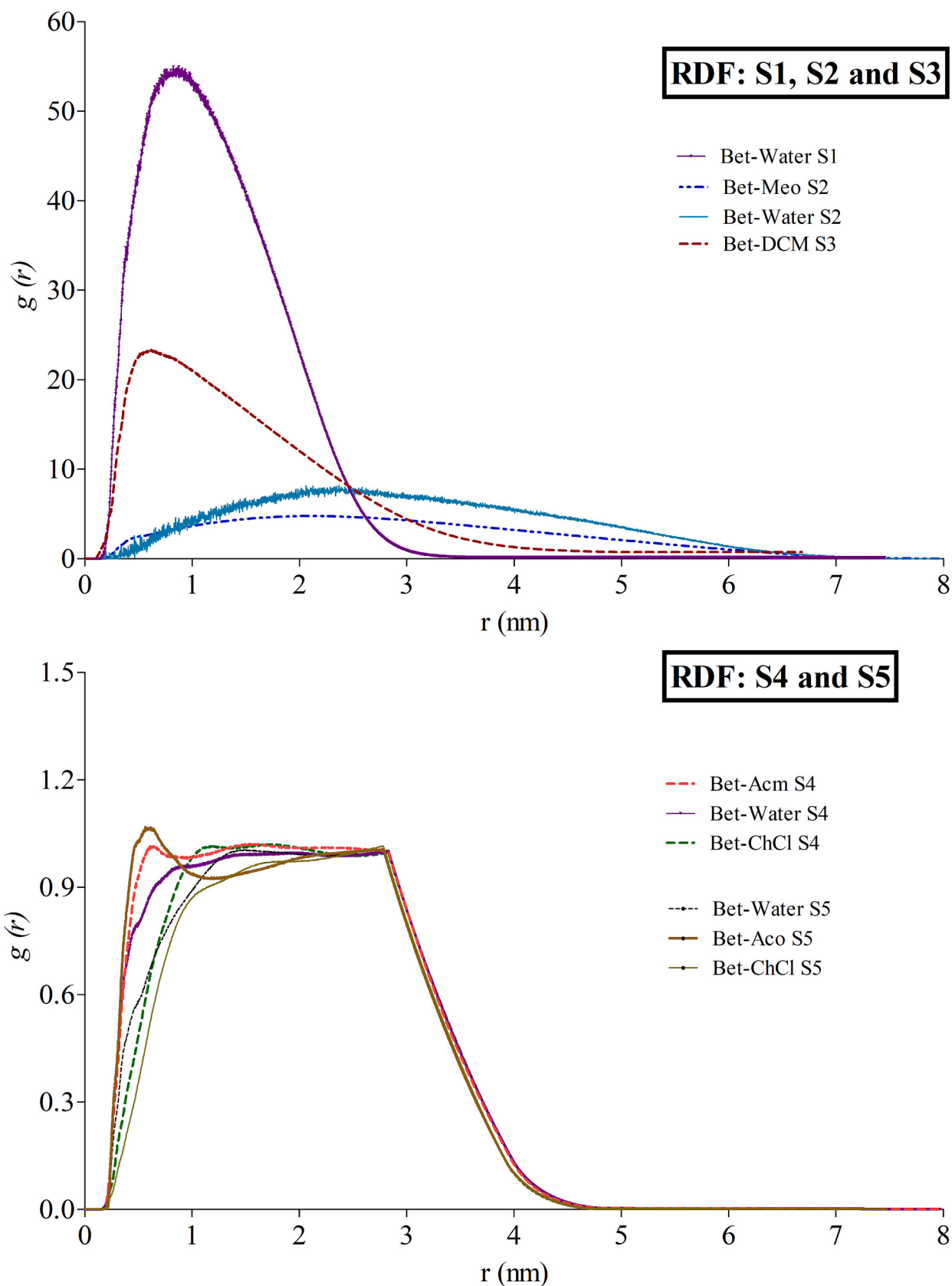


Fig. 5. Radial distribution functions (RDF) between beta-carotene and solvent components in Systems 1 to 5. The upper panel compares System 1: water (H_2O), System 2: a mixture of water (H_2O) and methanol (CH_3OH), and System 3: dichloromethane (DCM , CH_2Cl_2). The lower panel shows systems containing deep eutectic solvents: System 4 (choline chloride ($\text{C}_5\text{H}_{14}\text{ClNO}$), malic acid ($\text{C}_4\text{H}_6\text{O}_5$), and water) and System 5 (choline chloride, oxalic acid ($\text{C}_2\text{H}_2\text{O}_4$), and water). The $g(r)$ function is plotted for the interactions between beta-carotene and each solvent component.

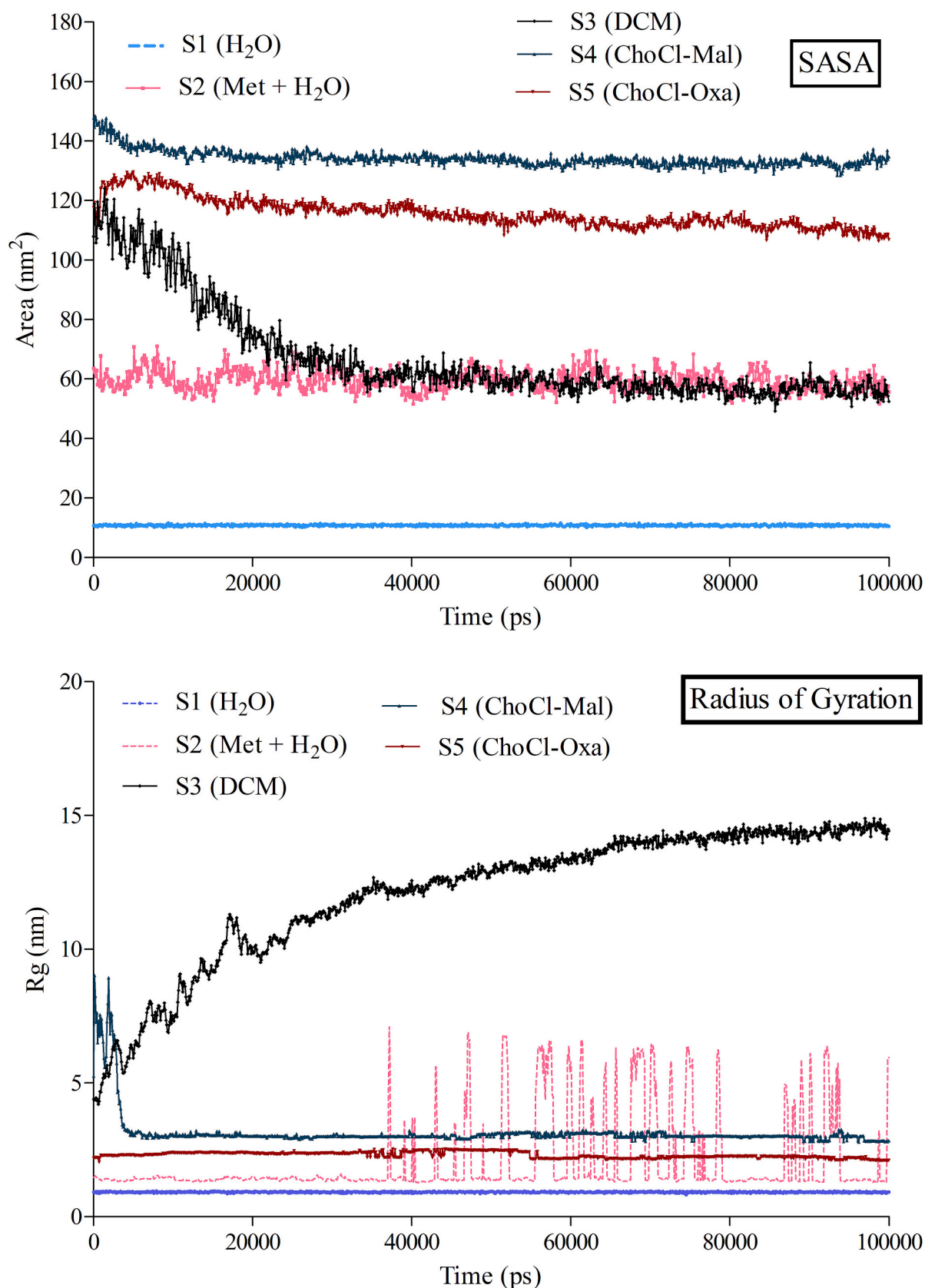


Fig. 6. Analysis of solvent-accessible surface area (SASA) and radius of gyration (Rg) of beta-carotene in Systems 1 to 5. The upper panel shows the temporal evolution of the solvent-accessible surface area (SASA) in the following systems: S1: water (H₂O), S2: mixture of methanol (CH₃OH) and water, S3: dichloromethane (DCM, CH₂Cl₂), S4: choline chloride (C₅H₁₄ClNO) with malic acid (C₄H₆O₅) and water, and S5: choline chloride with oxalic acid (C₂H₂O₄) and water. The lower panel presents the evolution of the radius of gyration (Rg) of beta-carotene, reflecting its degree of compactness and structural stability in each medium.

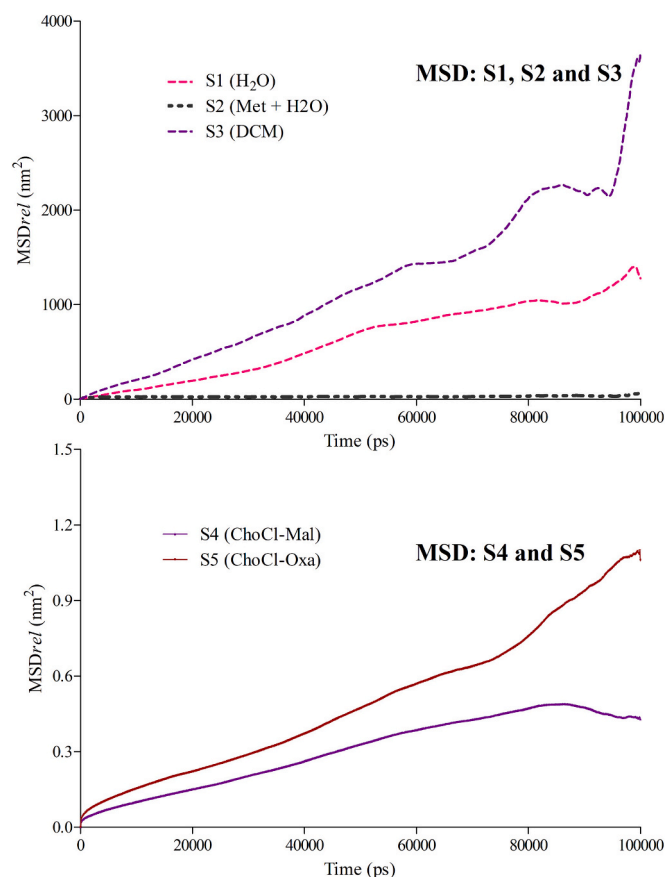


Fig. 7. Analysis of beta-carotene mobility through mean squared displacement (MSD) in Systems 1 to 5. The temporal evolution of the MSD of beta-carotene is shown for the following systems: S1: water (H_2O), S2: mixture of methanol (CH_3OH) and water (upper panel); S3: dichloromethane (DCM, CH_2Cl_2); S4: choline chloride ($\text{C}_5\text{H}_{14}\text{ClNO}$) with malic acid ($\text{C}_4\text{H}_6\text{O}_5$) and water, and S5: choline chloride with oxalic acid ($\text{C}_2\text{H}_2\text{O}_4$) and water (upper panel).

oxalic acid system (S5) provides a slightly more dynamic environment for β -carotene than the choline chloride/malic acid system (S4), possibly due to structural differences between the hydrogen bond donor acids that modulate the microstructure and viscosity of the medium.

In this study, the interaction of β -carotene with various solvent environments was investigated through molecular dynamics simulations to evaluate its structural and dynamic behavior, as well as its degree of solvation in both conventional solvents and natural deep eutectic solvents (NADES). Due to the pronounced hydrophobicity of β -carotene, the composition of the solvent medium plays a crucial role in determining its conformational stability, solubility, and practical applicability in extraction or formulation processes [29]. To address this, we analyzed key descriptors, including solvent-accessible surface area (SASA), radius of gyration (R_g), mean squared displacement (MSD), radial distribution functions (RDF), and specific molecular interactions.

The novelty of this work lies in the atomistic-level characterization of β -carotene behavior in NADES compared to conventional solvents, providing unprecedented insight into its solvation dynamics and structural organization. This comprehensive approach revealed clear compatibility patterns between the solute and solvent systems, offering rational criteria for designing customized solvation environments for lipophilic compounds. The simulation results align closely with experimental reports [30–32], which have shown that NADES not only improve the solubility of hydrophobic bioactives but also stabilize them through directional interactions like hydrogen bonding and van der Waals forces [33]. Notably, Bello et al. [31] emphasized the ability of NADES to form structured, biocompatible microenvironments, while

Bragagnolo et al. [32] reported enhanced carotenoid recovery and protection attributed to the polarity and low volatility of NADES.

Among conventional solvents, dichloromethane (DCM, S3) was included as a nonpolar organic benchmark, providing valuable contrast to NADES and aqueous media. DCM enabled remarkably high β -carotene mobility, as evidenced by the highest MSD values across all systems. However, this came at the cost of conformational destabilization. The increasing R_g and intermediate SASA values suggest unfolding and possible molecular disaggregation over time. RDF profiles confirmed poor spatial organization around the solute, reinforcing that while DCM enhances dispersion through strong dispersion forces, it fails to maintain conformational integrity.

This observation highlights a key point: solubility and stability are not always correlated. While DCM is effective in dissolving β -carotene, it behaves as a non-specific solvent incapable of sustaining the solute's structure over time. Compared to aqueous environments (S1 and S2), where either hydrophobic collapse (S1) or poorly organized partial solvation (S2) is observed, DCM offers better solubilization but lacks structural control. These findings underscore the limitations of conventional organic solvents in applications requiring both solubility and stability, particularly in pharmaceutical or nutraceutical systems.

In contrast, NADES-based systems—particularly choline chloride–malic acid (ChoCl-Mal, S4)—demonstrate a favorable balance between solvation and stabilization. S4 exhibited high SASA ($\sim 150 \text{ nm}^2$), intermediate R_g (3–4 nm), and flat RDF curves indicating a homogeneous solvation shell, with low MSD values reflecting the high viscosity of the medium rather than aggregation. Visualizations confirmed multiple hydrogen bonds between β -carotene and malic acid functional groups, supporting its stabilizing role. These findings are consistent with experimental evidence [9], which demonstrated the efficacy of NADES in solubilizing lipophilic antioxidants via extensive hydrogen bonding and high solvent–solute affinity.

System S5 (ChoCl-Oxa) showed a denser solvation shell and more compact solute conformation (SASA ~ 120 – 130 nm^2 ; $R_g \sim 2 \text{ nm}$), along with higher mobility (MSD $\approx 1.2 \text{ nm}^2$) than S4. This was attributed to the simpler molecular structure of oxalic acid, which allows for more dynamic motion while maintaining structural encapsulation. Hydrogen bonding networks in S5 appeared more geometrically organized than in S4, suggesting a distinct affinity pattern favorable for encapsulation. This structural behavior corroborates experimental observations [42] reporting increased extraction efficiency, thermal stability, and reduced oxidative degradation using NADES compared to conventional solvents.

Energetically, the solvation of β -carotene further illustrates these trends [34,35]. The weakest interactions were observed in water (S1), with minimal electrostatic and van der Waals contributions, resulting in the highest (least negative) total energy—indicative of poor stabilization [36]. The addition of methanol (S2) improved solvation energy through enhanced polarity and hydrogen bonding, reaching a significantly more negative total energy ($-110,685 \text{ kcal/mol}$). However, this mixture still lacked structural integrity and dynamic compatibility.

Among all systems, ChoCl-Mal (S4) exhibited the strongest total interaction energy ($-405,139 \text{ kcal/mol}$), with major contributions from electrostatic interactions ($-166,066 \text{ kcal/mol}$), suggesting synergistic stabilization via hydrogen bonding, π - π stacking, and ionic effects. This supports earlier findings that hydrophobic DES, especially those based on organic acids or fatty acids, can effectively extract carotenoids due to their specific interaction capabilities [37,38].

System S5 displayed the most intense electrostatic interactions ($-228,842 \text{ kcal/mol}$), but its total energy was less negative than that of S4, suggesting less favorable van der Waals or steric interactions. This confirms that beyond strong electrostatic stabilization, geometric compatibility and solvation shell architecture are crucial for overall solute stabilization [39,40].

The aqueous systems, S1 and S2, showed significant limitations. In S1, hydrophobic stacking, low SASA (~ 0), and minimal RDF structuring point to molecular collapse, despite high mobility (MSD $> 1400 \text{ nm}^2$),

likely due to water's low viscosity and lack of specific interactions. S2 revealed structural instability, heterogeneous solvation (SASA ~ 60 nm²), fluctuating Rg (~ 9 nm), and restricted diffusion (MSD ~ 10 nm²), suggesting poor compatibility and solvation performance. These findings are consistent with previous reports [41] showing that short-chain alcohols do not necessarily improve carotenoid solubility due to persistent microheterogeneity and dynamic instability.

Taken together, the results confirm that choline chloride-based NADES offer superior environments for the solubilization and stabilization of β -carotene. ChoCl-Mal enables extended conformations and high surface accessibility, while ChoCl-Oxa favors compact, stable encapsulation. Both systems surpass the performance of conventional solvents, including DCM, by maintaining directional interactions, dynamic stability, and low toxicity. These characteristics position NADES as chemically tunable platforms with broad potential in lipophilic compound delivery.

The combined analysis of structural (SASA, Rg), dynamic (MSD), and topological (RDF) metrics, along with energy calculations and trajectory visualizations, reveals the ability of NADES to enhance solubility while preserving structural integrity. This is of particular interest for improving the bioaccessibility of lipophilic nutrients, engineering nanocarriers, and developing advanced pharmaceutical formulations [43]. Recent studies have also demonstrated the effectiveness of NADES as therapeutic carriers, capable of improving solubility, permeability, and bioavailability through structured solvation environments and molecular-level interactions [44].

Based on these findings, ChoCl-Mal (S4) emerges as the most promising candidate for experimental validation in β -carotene extraction. Its ability to support extended conformations, uniform solvation, and strong hydrogen bonding suggests it can enhance both solubility and stability. Moreover, the sustainability, biocompatibility, and low environmental impact of NADES make them ideal for green extraction and formulation strategies in food, cosmetic, or pharmaceutical applications.

3.3. Functional and physicochemical insights into β -carotene solubilization in NADES

3.3.1. FTIR spectroscopy

FTIR spectra were obtained for β -carotene dissolved in the four solvent systems (S1–S5) (Fig. 8), with the aim of identifying changes in its characteristic vibrational modes as evidence of specific interactions with the surrounding solvent environment. In all systems, the main

absorption bands of β -carotene were clearly detected, including: ~ 960 cm⁻¹, attributed to torsional vibrations of conjugated C=C bonds; ~ 1160 cm⁻¹, corresponding to C—C stretching in the polyisoprenoid backbone; ~ 1455 cm⁻¹, associated with methyl group (—CH₃) bending vibrations; ~ 1520 cm⁻¹, characteristic of conjugated C=C stretching within the carotenoid chromophore; and ~ 1670 cm⁻¹, assigned to isolated C=C stretching, which is less conjugated than the former (Fig. 8).

In the aqueous system (S1), the β -carotene bands appeared poorly defined, with low intensity and slight overlap with solvent-related signals. This indicates limited interaction between β -carotene and water, consistent with its well-documented low solubility in polar protic solvents due to its hydrophobic polyene backbone [45]. The weak spectral definition suggests poor dispersion and minimal molecular affinity between the solute and the aqueous phase, which limits the applicability of such systems in aqueous formulations.

In the methanol–water mixture (S2), the characteristic bands became slightly more intense and better resolved, particularly the conjugated C=C band (~ 1520 cm⁻¹) (Fig. 8). This moderate enhancement reflects partial improvement in β -carotene dispersion, likely due to the amphiphilic nature of methanol, which can transiently stabilize hydrophobic molecules through dipole-induced interactions [46]. However, broad solvent bands between 3200 and 3500 cm⁻¹ persisted, contributing to spectral interference and indicating incomplete solute–solvent compatibility.

The FTIR spectrum of β -carotene in DCM (S3) revealed strong and well-defined absorption bands, particularly those corresponding to the conjugated C=C vibrations (~ 960 – 1520 cm⁻¹), with minimal spectral interference in the O—H region due to the aprotic nature of the solvent (Fig. 8). The band at ~ 2920 cm⁻¹, indicative of C—H stretching vibrations, was also clearly visible. These results confirm the efficient solvation of β -carotene in DCM, attributable to its non-polar character and structural compatibility with the polyene backbone. The clear and intense vibrational profile observed in DCM supports its role as a positive control in solubility and stabilization studies of hydrophobic compounds.

In contrast, the ChoCl-Mal system (S4) exhibited a pronounced increase in the intensity of β -carotene bands, especially in the 960–1520 cm⁻¹ region (Fig. 8). These bands were sharper and showed minimal shifts, suggesting more effective solvation and favorable interactions between the carotenoid and the NADES components. Additionally, a broad band at ~ 3400 cm⁻¹, attributed to O—H stretching, indicates the presence of an extensive hydrogen-bonding network—a hallmark of deep eutectic solvents (DES) that promotes molecular encapsulation and stabilization [9,47]. This structured microenvironment enhances solute–solvent affinity and stabilizes hydrophobic π -conjugated systems without disrupting their vibrational characteristics. These observations are consistent with recent studies reporting improved solubility and stability of carotenoids in choline chloride-based NADES, particularly when employing malic acid as the hydrogen bond donor [28].

The ChoCl-Oxa system (S5) also showed well-defined β -carotene bands, albeit with lower relative intensity than S3 (Fig. 8). Interference from solvent signals was less pronounced in the low-frequency region, and the O—H band near ~ 3400 cm⁻¹ was noticeably weaker, suggesting a less extensive hydrogen-bonding network. This may reflect a distinct solute–solvent interaction mechanism, potentially involving more compact molecular arrangements rather than extended hydrogen bonding. Oxalic acid, being a smaller and more acidic molecule, may result in higher viscosity and reduced solute accessibility, as reported in recent structure–property studies of NADES [48].

From a spectroscopic perspective, although DCM exhibited strong and well-resolved β -carotene bands with minimal interference—demonstrating excellent solvation capacity due to its non-polar nature—ChoCl-Mal (S3) clearly emerges as the most efficient green solvent among those tested. This is supported by the combination of high vibrational band intensities, minimized spectral overlap, and a well-

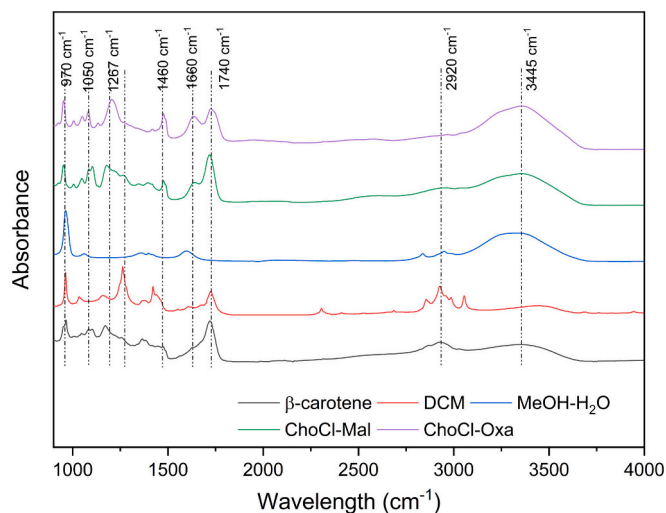


Fig. 8. FTIR spectra of β -carotene dissolved in five solvent systems: S1 (H₂O), S2 (methanol + H₂O), S3: dichloromethane (DCM, CH₂Cl₂), S4 (ChoCl-Mal), and S5 (ChoCl-Oxa). Characteristic vibrational bands associated with β -carotene and NADES components are highlighted.

defined O—H stretching region indicative of a robust hydrogen-bonding network. These features reflect a structured microenvironment that enhances solute–solvent affinity, stabilizing hydrophobic π -conjugated systems without altering their vibrational signature. Importantly, ChoCl–Mal offers a biocompatible and eco-friendly alternative to traditional organic solvents such as acetone or DCM, which are effective but present environmental and health hazards [48]. The enhanced solvation observed in S3 can be attributed to the synergistic interactions between choline chloride and malic acid, resulting in a tunable medium suitable for stabilizing hydrophobic bioactive compounds.

3.3.2. UV–vis absorption analysis

The UV–Vis absorption spectra of β -carotene dissolved in the four solvent systems—S2 (methanol:H₂O), S3 (DCM), S4 (ChoCl–Mal), and S5 (ChoCl–Oxa)—are shown in Fig. 9. All spectra display the characteristic vibronic fine structure of β -carotene, with a well-defined doublet centered around 450 nm, consistent with previous reports on carotenoids in solution [49]. This doublet corresponds to π – π^* electronic transitions within the extended conjugated polyene system and serves as a reliable indicator of chromophore integrity and molecular dispersion [50].

System S3 (DCM) exhibited the highest absorbance intensity (~ 1.10), with sharp and well-resolved vibronic bands and minimal baseline noise. These features indicate excellent solubilization and homogeneous dispersion of β -carotene, facilitated by the structural compatibility of the non-polar solvent with the polyene backbone. The clear vibronic structure confirms that DCM provides an efficient solvation environment that preserves the chromophore's electronic integrity. However, its classification as a volatile and non-green solvent limits its applicability in sustainable processes.

In system S2 (methanol:H₂O), the absorbance was slightly lower (~ 0.90), yet still prominent, suggesting effective but moderate solubilization. The absorption profile was broader, with higher background interference, which could indicate partial aggregation or dynamic instability due to the polar protic nature of the mixture. Methanol likely assists β -carotene dispersion through dipole-induced interactions, though it does not prevent some degree of molecular disorder [45].

System S4 (ChoCl–Mal) showed a maximum absorbance of approximately 0.70, along with sharp peaks and well-resolved vibronic bands. This reflects efficient molecular dispersion within the NADES matrix, likely due to hydrogen bonding and π – π interactions between the β -carotene and the solvent components. The spectral clarity and moderate intensity suggest ChoCl–Mal offers a good balance between

solubilization and chromophore stabilization, supported by FTIR and molecular dynamics data [28,47].

System S5 (ChoCl–Oxa) presented the lowest absorbance (~ 0.60), suggesting a reduced solubilization capacity. Nevertheless, the vibronic structure remained preserved, indicating that the electronic configuration of β -carotene is not disrupted. The attenuated signal may be attributed to the higher viscosity and stronger intermolecular interactions in the oxalic acid-based NADES, which can hinder molecular mobility and full solute dispersion [4].

Notably, no significant bathochromic or hypsochromic shifts were observed in the absorption maxima across the four systems, confirming that the electronic structure of β -carotene is preserved regardless of solvent polarity. This is particularly important for applications requiring retention of bioactivity linked to conjugation.

Among the green solvents, ChoCl–Mal stands out as the most promising system, offering a favorable combination of solubility, spectral definition, and molecular stability. This supports its potential application in eco-friendly carotenoid extraction and formulation processes for food, nutraceutical, and pharmaceutical uses.

3.3.3. Rheological behavior

The shear stress versus shear rate curves for β -carotene solutions in the different solvent systems (S2–S5) are presented in Fig. 10, highlighting significant differences in their rheological behavior. These variations directly influence both the efficiency of solute diffusion and the physicochemical stability of β -carotene in solution. The flow behavior of all four systems—S2 (methanol + water), S3 (DCM), S4 (ChoCl–Mal), and S5 (ChoCl–Oxa)—was accurately described by the Newtonian model, with coefficients of determination exceeding 0.996, mean squared errors approaching zero ($MSE \approx 0$), and highly significant p -values ($p < 0.01$). These results are consistent with the expected behavior of dilute, colloidal stable fluids lacking yield stress and viscoelastic properties, characteristic of homogeneous systems where the formation of extensive colloidal networks is suppressed [51].

From a physicochemical standpoint, the low flow resistance and constant viscosity—hallmarks of Newtonian fluids—facilitate mass transfer processes of β -carotene from the plant matrix into the extraction medium. This becomes particularly advantageous in extraction strategies assisted by ultrasound or mechanical agitation, where enhanced cellular disruption and boundary layer thinning promote solute diffusivity [51].

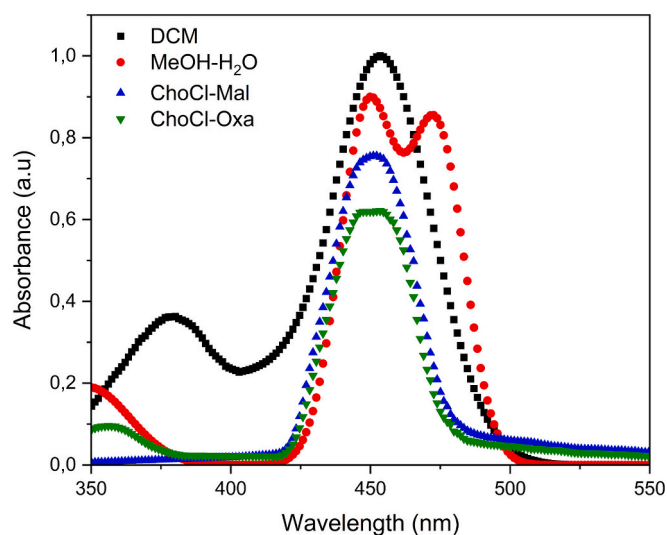


Fig. 9. UV–Vis absorption spectra of β -carotene in three solvent systems: S2 (methanol + H₂O), S3 (DCM), S4 (ChoCl–Mal), and S5 (ChoCl–Oxa).

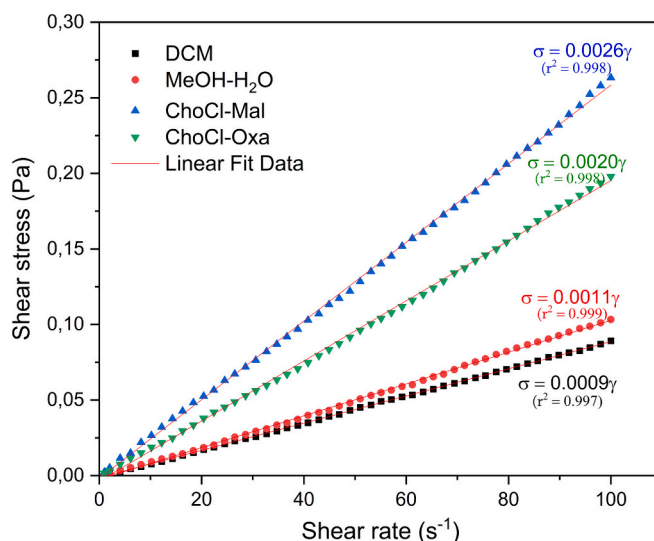


Fig. 10. Shear stress versus shear rate curves for β -carotene solutions in S2 (methanol + H₂O), S3 (DCM), S4 (ChoCl–Mal), and S5 (ChoCl–Oxa). Linear fits confirm Newtonian behavior in all systems, with different viscosity coefficients indicated by the slope ($\sigma = \eta\gamma$).

System S3 (DCM) exhibited the lowest viscosity ($\sigma = 0.0009 \gamma$), reflecting minimal flow resistance and high molecular mobility. This promotes rapid solute diffusion and efficient dispersion of β -carotene, consistent with its high solubility in non-polar environments. However, the absence of a structured hydrogen-bonding network in DCM limits its ability to prevent colloidal instability over time.

System S2 (methanol + water) showed slightly higher viscosity ($\sigma = 0.0011 \gamma$) while maintaining Newtonian behavior. This moderate viscosity facilitates diffusion but, due to its high polarity and lack of supramolecular structuring, the system may offer limited protection against aggregation or oxidative degradation—especially in the presence of dissolved oxygen [52,53].

Systems S4 (ChoCl–Mal) and S5 (ChoCl–Oxa) exhibited the highest viscosities among the systems studied ($\sigma = 0.0026 \gamma$ and $\sigma = 0.0020 \gamma$, respectively), yet still retained Newtonian flow profiles (Fig. 10). Rather than impeding diffusion, this moderate increase in viscosity suggests the presence of denser supramolecular networks capable of mitigating aggregation or sedimentation processes—an effect consistent with the moderate colloidal stability indicated by zeta potential measurements. This observation aligns with previous reports of enhanced physicochemical stability of β -carotene extracted using hydrophobic NADES from natural matrices such as pumpkin [54].

At the molecular level, β -carotene, being highly apolar, exhibits strong affinity for environments that facilitate hydrophobic and π - π stacking interactions. In this context, DCM (S3), due to its low polarity and high molecular mobility, offers a kinetically favorable medium for rapid β -carotene solubilization, as reflected in its high absorbance and minimal viscosity. Additionally, recent theoretical studies suggest that the formation of transient dicationic intermediates during solubilization in halogenated solvents may contribute to enhanced kinetic dispersion and reduced aggregation propensity, albeit temporarily [56]. However, the absence of supramolecular stabilizing networks—such as hydrogen bonding or sterically structured microdomains—limits DCM's capacity to ensure long-term colloidal and oxidative stability, particularly under industrial or storage conditions.

In contrast, NADES systems (S4 and S5), with their dynamic supramolecular architecture and partially apolar character, enable not only effective solubilization but also kinetic and thermodynamic stabilization of β -carotene. The combination of hydrogen bonding, Van der Waals forces, and moderate viscosity fosters the formation of protective microenvironments that mitigate aggregation, limit molecular mobility, and reduce oxidative degradation [55,56]. These properties, coupled with Newtonian flow behavior, highlight the potential of NADES as multifunctional solvents that balance solubility with structural stabilization.

Although system S2 enables rapid initial solute diffusion due to its low viscosity, it lacks the capacity to form protective supramolecular interactions. This limitation may compromise the physical and oxidative stability of β -carotene in aqueous polar media [54,55].

Taken together, these findings underscore the importance of simultaneously addressing mass transfer efficiency and physicochemical stability in the rational design of sustainable extraction systems. In this context, while DCM offers benchmark performance in terms of solubilization, NADES emerge as the most promising green alternative, capable of effectively balancing both aspects. This dual functionality not only enhances solute recovery but also contributes to preserving the structural and functional integrity of high-value lipophilic compounds such as β -carotene.

3.3.4. Zeta potential analysis

Fig. 11 displays the zeta potential values of β -carotene solutions in four solvent systems: S2 (methanol:H₂O), S3 (DCM), S4 (ChoCl–Mal), and S5 (ChoCl–Oxa). System S2 exhibited the highest zeta potential ($\sim +33$ mV), indicating strong electrostatic repulsion between dispersed particles. This value suggests a thermodynamically stable colloidal system with minimal aggregation tendency [57]. Such behavior is

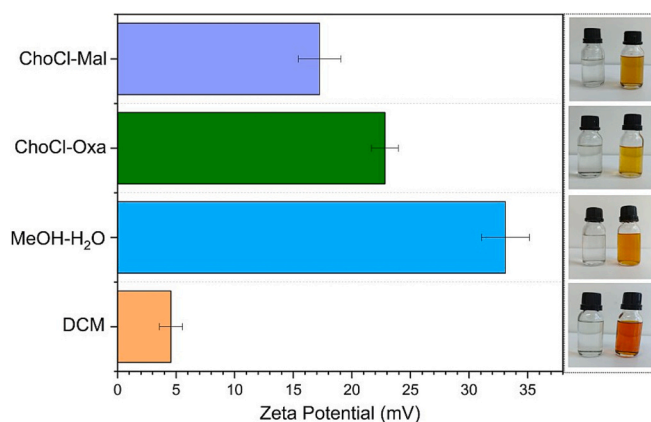


Fig. 11. Zeta potential values of β -carotene in S2 (methanol + H₂O), S3 (DCM), S4 (ChoCl–Mal), and S5 (ChoCl–Oxa). Error bars represent standard deviation. Photographs on the right show the visual appearance of each solvent system before and after carotenoid solubilization.

consistent with the high polarity of the hydroalcoholic medium, which promotes a uniform surface charge distribution and a well-developed electrical double layer [58]. However, despite its excellent dispersion performance, the use of methanol is restricted in food-related applications due to its toxicity.

System S3 (DCM) exhibited the lowest zeta potential ($\sim +6$ mV), indicating weak electrostatic stabilization. However, this low value should not be interpreted as indicative of poor solubilization capacity. On the contrary, DCM demonstrated the highest solubility for β -carotene, as evidenced by the well-defined absorbance bands in the UV–Vis spectra and clear vibrational features in FTIR analyses. This apparent contradiction can be explained by DCM's intrinsic physicochemical characteristics: as a non-polar, aprotic solvent with a very low dielectric constant and negligible hydrogen-bonding ability, DCM does not support the development of a structured electrical double layer around solute particles. Consequently, surface charge formation is minimal, leading to reduced electrostatic repulsion.

Moreover, DCM lacks ionizable functional groups, precluding the stabilization of β -carotene through solvation shell restructuring, as occurs in polar or protic solvents such as methanol or NADES. Importantly, recent theoretical studies suggest that the formation of transient dicationic intermediates during carotenoid solubilization in halogenated solvents like DCM may contribute to favorable kinetic behavior by temporarily reducing aggregation and enhancing dispersion [56]. Nonetheless, such kinetic advantages do not equate to sustained colloidal stability. In contrast, solvents like ChoCl–Mal, while exhibiting slightly lower solubilization efficiency, offer more favorable long-term stabilization due to their ability to form hydrogen-bonded networks, modulate viscosity, and support electrostatic interactions. These findings reinforce the need to consider both solubilization kinetics and colloidal stability in solvent system selection for β -carotene applications.

In contrast, systems S4 (ChoCl–Mal) and S5 (ChoCl–Oxa) showed intermediate zeta potential values ($\sim +25$ mV and $\sim +20$ mV, respectively), indicating moderate colloidal stability. Although these values do not exceed the classical ± 30 mV threshold for full colloidal stabilization, they are sufficient to delay aggregation and sedimentation. The higher zeta potential of ChoCl–Mal is attributed to malic acid's multiple functional groups, which promote stronger solute–solvent interactions and a more dynamic hydrogen-bonding network than oxalic acid. The molecular flexibility and asymmetry of malic acid likely favor the formation of partially charged solvation shells around β -carotene, enhancing particle stabilization.

Moreover, the relatively high viscosity of the NADES systems plays a complementary role by limiting Brownian motion and reducing the frequency of particle collisions. This kinetic stabilization contributes to

the maintenance of colloidal dispersion even in the absence of strong electrostatic repulsion. This dual mechanism—moderate zeta potential combined with rheological resistance—has been highlighted as a critical factor for ensuring the physical and chemical stability of carotenoids in natural eutectic media [54,59].

Importantly, although DCM exhibited a lower zeta potential ($\sim +15$ mV), which traditionally would suggest reduced colloidal stability, this characteristic actually aligns with its superior solubilization efficiency [60]. In industrial practice, such colloidal instability is not necessarily detrimental—instead, it often reflects rapid and complete solute dissolution. As noted by recent industrial reports, high solubility coupled with low kinetic resistance is preferred for carotenoid extraction processes where rapid phase transfer is required [61]. Moreover, the lower electrostatic repulsion may facilitate fast molecular reorganization and improve diffusivity, which are critical parameters under dynamic extraction conditions such as agitation or flow.

Additionally, the potential formation of dicationic intermediates during β -carotene solubilization in halogenated solvents like DCM could enhance kinetic stability by transiently reducing reactivity or aggregation [62]. This hypothesis, although not experimentally confirmed in this work, has been previously proposed as a contributing mechanism to the efficient encapsulation and solvent–solute compatibility observed in chlorinated systems. Therefore, while NADES systems like ChoCl–Mal provide a promising green alternative in terms of safety and tunability, DCM retains key kinetic advantages that are still relevant and appreciated in industrial settings. These insights further justify the need for complementary strategies that balance solubilization efficiency with sustainability.

4. Conclusion

This study demonstrates that choline chloride-based natural deep eutectic solvents (NADES), particularly the ChoCl–Mal system, offer a highly structured and functionally stable microenvironment that enhances the solubility, dispersion, and physicochemical protection of β -carotene. Molecular dynamics simulations revealed improved solvent-accessible surface area, reduced aggregation propensity, and favorable interaction energies between β -carotene and NADES components, especially in ChoCl–Mal. These computational predictions were strongly corroborated by experimental results: FTIR and UV–Vis spectroscopy confirmed effective solvation and preservation of the conjugated chromophore; rheological analysis revealed Newtonian behavior with moderate viscosity conducive to kinetic stabilization; and zeta potential measurements indicated sufficient electrostatic repulsion to maintain colloidal stability over time.

Although DCM exhibited the highest solubilization efficiency, enhanced kinetic behavior, and well-defined spectroscopic signals—making it highly attractive for industrial extraction processes—its low zeta potential and lack of a stabilizing supramolecular network may limit its utility in long-term storage or in food and pharmaceutical applications. By contrast, ChoCl–Mal demonstrated a balanced performance: while slightly inferior to DCM in terms of initial solubilization kinetics, it provided significantly better colloidal stabilization, molecular protection, and formulation potential.

These results highlight that the criteria for selecting solvents should not be limited to solubility alone. Instead, a comprehensive assessment—including colloidal stability, physicochemical protection, environmental safety, and regulatory compatibility—is essential. NADES, and particularly ChoCl–Mal, emerge as promising green solvents capable of meeting these multifaceted demands, offering a robust alternative to chlorinated solvents for the formulation and delivery of lipophilic bioactives.

Furthermore, considering the known sensitivity of NADES to water content, future research should systematically explore the influence of hydration on solvation behavior. Optimizing the water–NADES ratio may enable precise tuning of solvent properties, enhancing extraction

efficiency, bioactive stabilization, and delivery potential within green and scalable processing technologies.

CRediT authorship contribution statement

Yaremis Meriño-Cabrera: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Ivana Arrieta-Barba:** Writing – review & editing, Methodology, Investigation. **Luisa Florez-Vargas:** Methodology, Investigation. **Tiana Serna Fadul:** Methodology, Investigation. **Jorge Figueroa-Florez:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **José Severiche-Castro:** Methodology, Investigation, Formal analysis. **Jairo Salcedo Mendoza:** Writing – original draft, Supervision, Project administration, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the support of Minciencias, specifically through the Orquídeas: Mujeres en la Ciencia 2024 program, for granting the postdoctoral and young researcher fellowship. We also thank Minciencias for the financial support of the approved project No. 109622, contract 258-2024, executed by the Universidad de Sucre within the framework of the PADES research group.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2025.128251>.

Data availability

Data will be made available on request.

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